

## **Mapping NYC Crashes**

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## Which Domain?

For our project, we are seeking to find a relationship between vehicle collisions in NYC and available traffic control infrastructure. We are using a few data sources, but the purpose of these datasets can be traced to various entities interested in leveraging this data. The following entities are directly mentioned in the datasets we have.

The purpose of the data collection by NYC is to achieve the Vision Zero strategy, which aims to “eliminate traffic fatalities and severe injuries” (Vision Zero Communities). This program originates from “Sweden in 1997 [...] and since then has spread around the world. Reaching zero deaths requires the implementation of a Safe System approach. Road design and management should encourage safe speeds and manipulate appropriate crash angles to reduce injury severity” (*Zero deaths and safe system*).

Various entities utilize the Safe System Approach in relation to the Vision Zero organization. This approach identifies the key principles that drive the need for increased vigilance to create a system that maintains safety for all users of various modes of transportation.

New York State Department of Transit’s (NYSDOT) mission is “to provide a safe, reliable, equitable, and resilient transportation system that connects communities, enhances quality of life, protects the environment, and supports the economic well-being of New York State” (New York State Department of Transportation). USDOT is an administrative program that is led by the Secretary of Transportation. This department provides regulatory oversight for State DOT Administrations. The USDOT’s mission is “to deliver the world’s leading transportation system, serving the American people and economy through the safe, efficient, sustainable, and equitable movement of people and goods” (U.S. Department of Transportation Strategic Plan FY 2022).

Additionally, under the purview of our data we acknowledge the Traffic Signal Program Management which involves “organizing for the planning, maintenance, design, and operation of signalized intersections and traffic signal systems.” This department also falls under the DOT. Similarly, the United States Road Symbol Signs, under the DOT, are the standardization of road signs. “Familiarity with symbols on traffic signs is important for every road user in order to maintain the safety and efficiency of our transportation facilities.”

“On average there were 292 deaths each year due to motor vehicle traffic-related injuries among vehicle occupants, killing 1.6 of every 100,000 New Yorkers.” (Motor vehicle traffic crash statistics: New York State Residents). This statistic directly correlates to the need to continue pursuing the cause behind these accidents that result in death, even when a crash isn’t fatal, it could be the next time. “On average there were 106,612 emergency department(ED) visits each year due to motor vehicle traffic-related injuries among vehicle occupants” (Motor vehicle traffic crash statistics: New York State Residents).

The United States totals 40,990 people in 2023 lost their lives on roadways across the nation (The roadway safety problem), which equates to “the number of people that could fill the average professional baseball stadium”. This is not an insignificant number, while “roadway fatalities and the fatality rate declined consistently for 30 years” (The roadway safety problem). It is still not at the levels prescribed by Vision Zero. Thus, the need for improvement remains.

However, Vision Zero does have merit. “In New York City, Vision Zero has brought traffic deaths to historic lows and changed the culture on our streets through a comprehensive program of engineering, education, and enforcement” (Vision Zero, NYC Chapter). Citing fatalities down by 9% at the end of 2025. There are clear, actionable changes from the collected data; we would like to see the data come together into a model that drives action across multiple sources.

### **Which Data?**

For the main dataset we will be utilizing “Motor Vehicle Collisions – Crashes”. The metadata was last updated on March 22, 2026. This dataset is compiled by the city of New York; it is compiled by the Finest Online Records Management System (FORMS). Other previous methods are mentioned on data.gov, however FORMS is the latest to be used. “FORMS enables the police officers to electronically, using a Department cellphone or computer, enter all of the MV-104AN data fields and stores all of the MV-104AN data fields in the Department’s crime data warehouse. Since all of the MV-104AN data fields are now stored for each traffic collision, detailed traffic safety analyses can be conducted as applicable” (Publisher data.cityofnewyork.us).

Data Source 1 “Target Data” (NYC Motor Vehicle Crashes):  
<https://catalog.data.gov/dataset/motor-vehicle-collisions-crashes>

The second data source we are leveraging is “Traffic Signal and All-Way Stop Study Requests” compiled by the Department of Transit (DOT) in “service studies related to the traffic signals and their maintenance” (). This will be used to compare Vehicular crashes in relation to any requests within the vicinity for traffic signals or signage.

Data Source 2 “Request/Condition Data” (Traffic Signal and Stop Requests):  
[https://data.cityofnewyork.us/Transportation/Traffic-Signal-and-All-Way-Stop-Study-Requests/w76s-c5u4/about\\_data](https://data.cityofnewyork.us/Transportation/Traffic-Signal-and-All-Way-Stop-Study-Requests/w76s-c5u4/about_data)

The third data source is to show any changes in traffic signal conditions that have been reported in the area, and we want to relate the dates around the crash data. To see if there was a potentially down signal in the area that could be related to crashes.

#### Data Source 3 “Signage Data” (Traffic Signal Condition):

[https://data.cityofnewyork.us/Social-Services/Traffic-Signal-Condition\\_4/e5mv-wy8f/about\\_data](https://data.cityofnewyork.us/Social-Services/Traffic-Signal-Condition_4/e5mv-wy8f/about_data)

Our fourth data set includes all traffic signals maintained by the NYS DOT for points of reference with crashes, requests, and conditions.

#### Data Source 4 “Signal Data” (NYS DOT Maintained Traffic Signals):

[https://data.gis.ny.gov/datasets/c61a41873d4f4b06bae79c23ad98b9c6\\_0/about](https://data.gis.ny.gov/datasets/c61a41873d4f4b06bae79c23ad98b9c6_0/about)

### **Research Questions? Benefits? Why Analyze these data?**

This dataset can be leveraged in a variety of ways. The initial need to collect this data allows us to glean a lot of information. One way we would like to explore this dataset is to view traffic incidents and road signage or signals in the area where the incidents occur. Then we could see if these areas have no signage or signal. We would like to see if there are any requests for updates to the area. If there is or is not a request, we can identify areas in need of support. This could help eliminate traffic incidents and further the Vision Zero strategy for the city of New York.

Questions we wish to explore:

- Where are intersections with a high-risk of crashes? (More than 3 crashes in a year)
- What contributing factors are the most common at the high-risk crashes?
- Does the intersection have traffic signals?
- If not, is there a request for a traffic signal?
- If yes, would it be pertinent to expedite the traffic signal (for identifying the high-risk crash areas).

### **What Method?**

First, we will clean our data. We have a variety of datasets we are looking to combine to leverage our inquiries. We want to use our core dataset for the crash info; we will narrow our focus into a small portion of the data for a few years of data. If any coordinates are missing, we will exclude those data points for our purposes as we don’t want to assume any data for our model. We will standardize date and time formats as well as geolocation by converting LAT/LON into feet using EPSG:2263.

Next, we will work towards feature engineering for our dataset where we identify if a traffic signal/sign is found in the crash area or if there is a submission for a traffic signal/sign in the area already.

Lastly, we want to create a model that can identify risks with intersections. At this time, we will want to ensure our model (Potentially: Classification Model to easily translate our results to business entities) is trained with our data but also balancing this training data to ensure it isn't guessing incorrectly. We can check our model to ensure it is properly identifying risks.

### **Potential Issues?**

We can certainly run into a variety of issues with our model. We could have trouble getting all the data sets to work together and may have to pivot during the project to rework our assumptions coming in. The point of these projects is to give us an idea of working together and the flow of work, we understand that these tasks will not just be linear in completion but also, we will regress at some points and redo portions of this at times. The problem with this is that we have a limited short time period and will have to make concessions at times and be open and honest with ourselves about our skill level and comfort. To overcome these issues, we need to be flexible and cooperative in our group and keep the lines of communication open.

### **Concluding Remarks.**

This project seeks to work with the Motor Vehicle Collisions in New York City to evaluate where high-risk intersections exist and whether there is an issue with traffic signage in the areas. Our goal is to create a model that will help pull multiple datasets together to accomplish this task. The assumptions we have are that we will positively identify high-risk zones from the dataset that will prove there is a lack of signage that contributes to these accidents.

## References

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